

PROJECT DEVELOPMENT PLAN

PS 26190 – Secure Digital Document Management System for Legal and Investigation Documents

Existing Project: SecureCloudStorage

Repository: [GitHub – SecureCloudStorage](#)

Problem Statement ID: 26190

Development Approach: Phase-wise

Resource Policy: 100% Free / Open Source

Version Control: Git + GitHub

Primary Development Branch: feature/ps-26190-dms

1. Project Vision

The existing SecureCloudStorage project will be extended into a Secure Digital Document Management System (DMS) specifically designed for legal and investigation-related documents.

The system will provide a secure platform for:

- Case management
- Legal document storage
- Document organization and retrieval
- Role-based access control
- Document integrity verification
- Version control
- Audit logging
- Secure document sharing
- Evidence and chain-of-custody management
- Digital signatures
- OCR and intelligent search
- Security monitoring
- Backup and recovery

The project will be developed incrementally so that each phase produces a stable and demonstrable improvement to the existing system.

2. NON-NEGOTIABLE PROJECT CONSTRAINTS

These rules apply to the entire project and must not be violated unless the entire team agrees to formally change them.

Constraint 1 — Free Resources Only

All technologies, libraries, APIs, hosting platforms, databases, AI models, development tools and services used in the project must be:

- Free

- **Open source**
- **Free-tier without mandatory payment**
- **Locally runnable where possible**

We will NOT use:

- **Paid APIs**
- **Paid AI models**
- **Premium SaaS services**
- **Paid hosting**
- **Paid databases**
- **Paid plugins**
- **Services requiring a credit card/payment for essential functionality**

Important: Free-tier availability can change. Before adopting any external service, its current pricing and limitations must be verified.

If a required service becomes paid:

We replace it instead of paying for it.

3. DATA PRIVACY RULE

Because the system is designed for sensitive legal and investigation documents:

During development we will use ONLY:

- **Synthetic FIRs**
- **Dummy police reports**
- **Sample witness statements**
- **Artificial evidence records**
- **Test PDFs/images**
- **Fake user accounts**

We will NOT upload:

- **Real FIRs**
- **Real investigation documents**
- **Real personal information**
- **Real evidence**
- **Confidential court documents**
- **Sensitive government data**

to development or testing environments.

4. VERSION CONTROL POLICY

Version control will be treated as a mandatory project requirement, not an optional practice.

Main Branch

main

The main branch must always contain a stable, working version.

Nobody should directly develop on main.

5. DEVELOPMENT BRANCH

A dedicated branch will be created from the current stable project:

feature/ps-26190-dms

All PS-26190 development will initially happen through this branch.

6. FEATURE BRANCH STRUCTURE

Individual team members should create branches from the development branch.

Examples:

feature/case-management

feature/rbac

feature/audit-log

feature/document-versioning

feature/evidence-management

feature/digital-signature

feature/ocr-search

feature/security-dashboard

For bug fixes:

fix/login-error

fix/file-upload

fix/access-control

For security-related changes:

security/file-validation

security/session-management

security/encryption

7. COMMIT POLICY

Every commit must represent a logical and identifiable change.

Good:

feat: add case creation module

feat: add role based access control

fix: prevent unauthorized document download

security: validate uploaded file types

docs: update project architecture

test: add audit log tests

Avoid:

update

changes

final

final2

new

working

asdf

8. PULL REQUEST POLICY

No feature should be merged directly without review.

Required workflow:

Issue



Feature Branch



Development



Testing



Security Review



Pull Request



Teammate Review



Approval



Merge



Version Tag

9. PROHIBITED GIT CONTENT

The repository must NEVER contain:

- **.env files containing secrets**
- **API keys**
- **Passwords**
- **Database credentials**
- **Private keys**
- **Authentication secrets**
- **Real legal documents**
- **Real evidence**
- **Sensitive personal information**

A proper .gitignore must be maintained throughout development.

10. PROJECT PHASES

The project will be implemented through the following milestones.

PHASE 0 — BASELINE & PROJECT GOVERNANCE

Objective

Secure the existing project before making major changes.

Tasks

- 1. Freeze the current working version.**
- 2. Verify the current SecureCloudStorage application.**
- 3. Create the PS-26190 development branch.**
- 4. Document the existing architecture.**
- 5. Document the current database structure.**
- 6. Document existing encryption implementation.**
- 7. Document existing authentication.**
- 8. Record dependencies.**
- 9. Review .gitignore.**

10. Verify that no secrets exist in Git history/current files.

11. Create GitHub Issues/Milestones.

12. Create project development documentation.

Required Documents

README.md

CHANGELOG.md

DEVELOPMENT_LOG.md

TECHNOLOGY_DECISIONS.md

SECURITY.md

Deliverable

Stable and documented baseline of SecureCloudStorage.

Version

v0.1-foundation

PHASE 1 — CORE DOCUMENT MANAGEMENT & CASE MANAGEMENT

Objective

Transform the existing file-storage model into a case-centric document system.

Case Management

Implement:

- **Create Case**
- **View Case**
- **Update Case**
- **Case ID**
- **Case title**
- **Case description**
- **Department**
- **Assigned officer**
- **Case status**
- **Creation date**
- **Closing date**

Example

CASE-2026-001

Title:

Robbery Investigation

Status:

Under Investigation

Assigned Officer:

Officer A

Document Categories

Support:

- **FIR**
- **Police Report**
- **Witness Statement**
- **Charge Sheet**
- **Court Filing**
- **Evidence**
- **Forensic Report**
- **Legal Notice**
- **Judgment**
- **Other**

Document Metadata

Every document should maintain:

- **Document ID**
- **File name**
- **Case ID**
- **Category**
- **Uploaded by**
- **Upload date/time**
- **Version**
- **File hash**
- **Status**
- **Encryption status**

Deliverable

Case-centric secure document repository.

Version

PHASE 2 — ROLE-BASED ACCESS CONTROL

Objective

Ensure that users can access only information they are authorized to access.

Proposed Roles

Role	Primary Responsibility
Admin	System management
Investigating Officer	Case investigation
Police Officer	Authorized case operations
Legal Officer/Prosecutor	Legal documents
Forensic Officer	Forensic/evidence documents
Authority	Authorized review

Security Model

Use:

Deny by default → Allow only when authorized

Implement:

- **Role-based permissions**
- **Case-level authorization**
- **Document-level authorization**
- **View permission**
- **Upload permission**
- **Download permission**
- **Management permission**

Deliverable

Verified least-privilege access system.

Version

v0.3-rbac

PHASE 3 — AUDIT TRAIL & DOCUMENT INTEGRITY

Objective

Make every important document action traceable.

Audit Events

Record:

- **Login**
- **Logout**
- **Upload**
- **View**
- **Download**
- **Update**
- **Delete**
- **Share**
- **Permission change**
- **Case assignment**
- **Failed login**
- **Unauthorized access attempt**

Each record should contain:

User

Action

Document/Case

Timestamp

Result

Integrity Protection

Generate a SHA-256 hash for every important document.

Document

↓

SHA-256

↓

Stored Hash

When required:

Current File Hash

↓

Compare

↓

Original Hash

↓

MATCH / MISMATCH

Deliverable

Auditable and integrity-verifiable document system.

Version

v0.4-audit-integrity

PHASE 4 — DOCUMENT VERSION CONTROL & LIFECYCLE

Objective

Prevent uncontrolled modification of important documents.

Instead of:

ChargeSheet.pdf

being silently overwritten:

ChargeSheet

|— Version 1

|— Version 2

|— Version 3 ← Current

Maintain:

- **Version number**
- **Previous version**
- **Current version**
- **Modified by**
- **Modification timestamp**
- **Change description**

Document Lifecycle

Draft

↓

Reviewed

↓

Approved

↓

Archived

Where appropriate, implement:

- **Restore previous version**
- **Prevent unauthorized deletion**
- **Retention information**

- Legal hold

Deliverable

Controlled document lifecycle and version history.

Version

v0.5-document-lifecycle

PHASE 5 — EVIDENCE MANAGEMENT & CHAIN OF CUSTODY

Objective

Extend the system from documents to investigation evidence.

Evidence Record

Each evidence item should have:

Evidence ID

Case ID

Evidence Type

Description

Collected By

Collection Location

Collection Date/Time

Current Holder

Status

Hash

Chain of Custody

Track:

Collected

↓

Uploaded

↓

Transferred

↓

Examined

↓

Stored

↓

Presented

Every transfer must create an audit event.

Deliverable

Evidence lifecycle and chain-of-custody prototype.

Version

v0.6-evidence

PHASE 6 — DIGITAL SIGNATURES

Objective

Provide a mechanism for verifying the authenticity and integrity of important documents.

Prefer free/open-source cryptographic implementations.

Possible workflow:

Document

↓

Hash

↓

Digital Signature

↓

Signed Record

Store:

- **Signer**
- **Timestamp**
- **Document hash**
- **Signature**
- **Verification status**

Provide:

Verify Signature

Important Restriction

The system must not claim statutory/legal validity merely because a digital-signature feature has been implemented.

The project should clearly distinguish:

Technical signature verification

from

Legally recognized digital signature.

Deliverable

Digital signature and verification prototype.

Version

v0.7-digital-signatures

PHASE 7 — OCR & INTELLIGENT DOCUMENT SEARCH

Objective

Allow users to find information quickly, even inside scanned documents.

OCR

Use free/open-source technology such as:

- **Tesseract**
- **PaddleOCR**

Workflow:

Scanned Document



OCR



Extracted Text



Search Index

The original document remains preserved.

Search

Support:

- **Filename**
- **Case ID**
- **Document category**
- **Officer**
- **Date**
- **Metadata**
- **Extracted text**

Advanced Search

Example:

Find forensic reports associated with CASE-2026-001.

Optional Semantic Search

If resources permit:

Document

↓

Text

↓

Embedding

↓

Vector Search

↓

Relevant Documents

Only free/local solutions should be considered.

Deliverable

Fast and privacy-conscious document retrieval.

Version

v0.8-ocr-search

PHASE 8 — COLLABORATION & SECURITY MONITORING

Objective

Improve cooperation between authorized stakeholders.

Implement:

- **Case assignment**
- **Document sharing**
- **Permission management**
- **Notifications**
- **Security events**
- **Failed login monitoring**
- **Unauthorized access monitoring**

Security Dashboard

Example:

Security Events

Failed Logins: 12

Unauthorized Attempts: 3

Integrity Alerts: 0

Active Users: 8

Optional

- MFA
- Rate limiting
- Suspicious activity detection

Deliverable

Collaborative and security-aware DMS.

Version

v0.9-security-monitoring

PHASE 9 — BACKUP, RELIABILITY & DEPLOYMENT HARDENING**Objective**

Make the system reliable enough for final demonstration.

Implement/test:

- Database backup
- Encrypted backup
- Restore procedure
- File backup strategy
- Input validation
- File validation
- Upload restrictions
- Secure sessions
- Security headers
- Dependency review
- Error handling

Critical Rule

A backup feature is not considered complete until restoration has been tested.

Deployment

Only free deployment resources may be used.

If an existing deployment service becomes unsuitable or paid:

Find a free alternative or redesign the feature.

Deliverable

Hardened and recoverable system.

Version

v1.0-hardened

PHASE 10 — FINAL TESTING & PS COMPLIANCE

Objective

Verify that the final system actually satisfies PS 26190.

Testing must include:

Functional Testing

- **Registration/login**
- **Case creation**
- **Document upload**
- **Document retrieval**
- **Document sharing**
- **Versioning**
- **Evidence management**

Security Testing

- **Unauthorized access**
- **Invalid permissions**
- **Invalid file upload**
- **Tampered document**
- **Failed login**
- **Session security**
- **Access-control bypass attempts**

Integrity Testing

Original Document

↓

Hash

↓

Modified Document

↓

Hash

↓

Mismatch Detected

Final PS Mapping

Every requirement from PS 26190 must be mapped to:

Requirement

↓
Implemented Feature
↓
Testing Evidence
↓
Demo Evidence
Deliverable
Final PS-26190-compliant demonstration system.
Version
v1.1-ps-26190-final

11. OVERALL ROADMAP

Milestone Phase		Main Output
M0	Baseline	Git governance + stable foundation
M1	Case Management	Case-centric DMS
M2	RBAC	Secure authorization
M3	Audit + Integrity	Traceability + tamper detection
M4	Version Control	Document lifecycle
M5	Evidence	Chain of custody
M6	Digital Signatures	Signature verification
M7	OCR + Search	Intelligent retrieval
M8	Collaboration	Notifications + monitoring
M9	Hardening	Backup + security
M10	Final Testing	PS compliance + final demo

12. TECHNOLOGY SELECTION POLICY

The project will prefer technologies already used in SecureCloudStorage to minimize unnecessary migration.

Current Foundation

Frontend

HTML + CSS + JavaScript

Backend

Python + Flask

ORM

SQLAlchemy

Database

PostgreSQL / existing project database

Encryption

AES-256-GCM

Key Derivation

PBKDF2-HMAC-SHA256

Version Control

Git + GitHub

Possible Free/Open-Source Additions

Testing

pytest

OCR

Tesseract / PaddleOCR

Security Testing

OWASP ZAP

Hashing

SHA-256

Local AI

Open-source/local models where feasible

Vector Search

Free/open-source local solution

No technology should be added simply because it is popular.

Every new technology must answer:

1. Is it free?
2. Is it open source or genuinely free to use?
3. Does it introduce a mandatory paid dependency?
4. Does it handle sensitive data?
5. Can we run it locally?
6. Is there a free fallback?
7. Does the team actually need it?

13. TECHNOLOGY DECISION LOG

Every external technology should be recorded in:

TECHNOLOGY_DECISIONS.md

Use this format:

Technology	Purpose	Cost	License/Free Status	Privacy Risk	Alternative
Example	OCR	Free	Open Source	Low/Local	Alternative
Example	Database	Free tier	Verify	Medium	Local PostgreSQL

This prevents the team from accidentally building a feature that later becomes dependent on a paid service.

14. DEFINITION OF DONE

A feature is NOT complete merely because the code works on one developer's computer.

A feature is considered complete only when:

- ☐ Code implemented
 - ☐ Functional testing completed
 - ☐ Security testing completed where applicable
 - ☐ No known critical errors
 - ☐ Documentation updated
 - ☐ Git commit created
 - ☐ Pull request created
 - ☐ Teammate review completed
 - ☐ PR merged
 - ☐ Version updated/tagged where appropriate
 - ☐ Demonstrated successfully
-

15. TEAM DEVELOPMENT WORKFLOW

Every team member must follow:

SELECT ISSUE



CREATE BRANCH



IMPLEMENT FEATURE



TEST



SECURITY CHECK



COMMIT



PUSH



CREATE PULL REQUEST



TEAMMATE REVIEW



FIX REVIEW COMMENTS



APPROVE



MERGE



UPDATE DOCUMENTATION

16. PRIORITY RULE

The project must follow this order:

Priority 1 — Security

Encryption, authentication, authorization, integrity.

Priority 2 — Correctness

Cases, documents, evidence, audit logs.

Priority 3 — Reliability

Versioning, backup, recovery, error handling.

Priority 4 — Usability

Dashboard, search, collaboration, notifications.

Priority 5 — Intelligence

OCR, semantic search, AI summarization.

AI must never be implemented at the expense of core security and reliability.

17. FEATURE PRIORITY MATRIX

Feature	Priority	Status
Case Management	● Critical	Planned
Document Metadata	● Critical	Planned
RBAC	● Critical	Planned
Audit Trail	● Critical	Planned
Integrity Verification	● Critical	Planned
Version Control	● Critical	Planned
Secure Sharing	● Critical	Planned
Evidence Management	● High	Planned
Chain of Custody	● High	Planned
Digital Signatures	● High	Planned
OCR	● High	Planned
Advanced Search	● High	Planned
Security Dashboard	● High	Planned
Notifications	● Medium	Planned
MFA	● Medium	Planned
AI Semantic Search	● Advanced	Planned
AI Summarization	● Advanced	Planned
Blockchain/Immutable Ledger	● Experimental	Optional

Feature	Priority	Status
Retention/Legal Hold	● Advanced	Optional

18. GOLDEN RULES OF THIS PROJECT

Rule 1

No paid resources.

Rule 2

No direct development on main.

Rule 3

No secrets in GitHub.

Rule 4

No real sensitive legal data during development.

Rule 5

No feature is considered complete without testing.

Rule 6

Security features require negative testing.

Rule 7

Never claim an unimplemented feature.

Rule 8

Every major phase must produce a working version.

Rule 9

If a dependency becomes paid, replace it.

Rule 10

Security and correctness come before AI or flashy features.

19. FINAL PROJECT TARGET

At completion, SecureCloudStorage should evolve from:

A secure cloud file-storage application

into:

A secure, case-centric digital document management platform for legal and investigation documents with encryption, access control, document integrity, versioning, auditing, evidence management, secure collaboration, and intelligent document retrieval.

The final system should demonstrate:

Secure Storage + Case Management + Access Control + Integrity + Auditability + Evidence Tracking + Intelligent Retrieval

while remaining free-resource based, version-controlled, privacy-conscious, and technically demonstrable.

20. SINGLE SOURCE OF TRUTH

This document will be treated as the team's Project Development Master Plan.

Whenever a new feature is proposed, the team must first determine:

Which phase does it belong to?

Does it violate the free-resource constraint?

Does it introduce a security/privacy risk?

Can it be tested and maintained?

Does it contribute to PS 26190?

If the answer is unclear, the feature should not be added immediately. It should first be discussed and recorded in the project's issue tracker/technology decision log.

Development principle:

Build → Test → Review → Secure → Document → Merge → Tag → Move to the next phase.